Directorate of Curriculum and Teacher Education
Khyber Pakhtunkhwa, Abbottabad.
Induction Program Phase-III. (Exam-II)
PST

Time: 03hrs **Subject: Teaching of Mathematics** **Total Marks: 70**

Section A **(Marks: 14)**

Q.1. Choose the correct answer and write them in your Answer book.

1. Identify the set of prime Numbers:
   (a) {1, 3, 5, 7, 8, ...} (b) {2, 4, 6, 8, 10, ...}
   (c) {2, 3, 5, 7, 9, ...} (d) {1, 2, 3, 4, 5, 7, ...}
2. A set of whole number is represented by:
   (a) N (b) W (c) Z (d) O
3. The set of equivalent friction of $\frac{2}{7}$ is:
   (a) {4/14, 6/21, 8/28...} (b) {2/7, 3/14, 4/28 ...}
   (b) (c) {1/7, 3/14, 5/28...} (d) {4/14, 6/21, 8/28...}
4. The LCM of 6 and 8 is:
   (a) 6 (b) 8 (c) 24 (d) 48
5. Student learning outcome (SLO) focus on enhancing:
   (a) Course (b) Memory (c) Concept (d) Grade level
6. The sum of any two whole numbers is a whole number, this property is referred as:
   (a) commutative property (b) Associative property
   (c) Distributive property (d) Closure property
7. The mathematics is:
   (a) Ornamental (b) Difficult (c) Logical (d) Unsystematic
8. While teaching geometry, the teaching method used is referred as:
   (a) Lecture method (b) Analytical method
   (b) Problem solving method (d) Heuristic Method
9. The angle of measure of 270°
   (a) Acute angle (b) Obtuse angle (c) Reflex angle (d) Straight angle
10. The number whose 7% is 42 is:
    (a) 300 (b) 400 (c) 500 (d) 600

    $\frac{7}{100} \times 42$

    $7\% \times x = 42$

    $x = \frac{\overset{6}{42} \times 100}{7}$

    $x = 600$
11. The diameter of a circle whose radius is 7 cm is:
    (a) 3.5 cm (b) 7 cm (c) 14 cm (d) 21 cm
12. Which is not the co-curricular activity:
    (a) Mathematics Exhibition (b) Mathematics Fair
    (c) Mathematics Club (d) Classroom Teaching
13. The statement that indicates interrelationship between two or more concepts is known as:
    (a) Principle (b) Fact (c) Theorem (d) Rule
14. The sum of measures of all the four angles of a quadrilateral is:
    (a) 90° (b) 180° (c) 360° (d) 720°

Section B

(Marks: 28)

NOTE: Attempt any Four questions. All questions carry equal marks.

Q.2. Differentiate between Two dimensional and Three dimensional shapes. Enlist few Two dimensional and Three dimensional shapes.

Q.3. Define the project method of teaching mathematics and enlist its steps.

Q.4. What are the set squares in geometry box, describe their forms and angular description.

Q.5. The total age of three kids in a family is 27 years. What will be their total age after 3 years?

Q.6. Simplify $133 - 19x^2 + 5$

Q.7. Find the HCF of 60, 75 and 105.

Q.8. Simplify $[36 \div (-9)] \div [(-24) \div 6]$

Section C

(Marks: 28)

NOTE: Attempt any Two questions.

Q.9. Why mathematics education is important in primary school. Explain the significance of place of mathematics and its need at primary school curriculum.

Q.10. Write the names of different methods used in teaching of Mathematics and explain analytic method with examples.

Q.11. Explain the importance of lesson planning. Write a comprehensive lesson-plan for teaching of 3D shapes.

Q.12. What is the area of circle in terms of π whose diameter is 16cm.

Q.13. Write note on any three of the following teaching approaches for teaching of mathematics mentioning them advantages and disadvantages.

i. Constructive Approach

ii. Montessori Approach

iii. Cooperative Approach

iv. Project approach